
Session 4 — Analysis II

Exercise sheet

Functions with several inputs, functions with several outputs, and what happens when you write a differential equation for each of them at once. Two examples recur: the squared-error surface of a fitted model, whose bottom is found by following the gradient, and a two-accumulator decision model, whose behaviour is decided by the eigenvalues of a 2×2 matrix. Show your reasoning; most answers are short.

Exercise 1 Partial derivatives. Compute $\partial f/\partial x$ and $\partial f/\partial y$.

- (a) $f(x, y) = x^2 + 3xy + y$;
 - (b) $f(x, y) = x^2 + 4y^2$;
 - (c) $f(x, y) = x^2y + \sin y$;
 - (d) $f(x, y) = e^{-(x^2+y^2)}$.
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Exercise 2

- (a) For $f(x, y) = x^2 + 3xy + y$, evaluate $\partial f/\partial x$ at $(1, 0)$ and at $(1, 2)$.
 - (b) In one sentence: why are the two answers different, and what does that tell you about functions of several variables?
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Exercise 3 Contours and the gradient.

- (a) Sketch the contour lines $f = 1$ and $f = 4$ of $f(x, y) = x^2 + 4y^2$.
 - (b) Compute $\nabla f(1, 1)$.
 - (c) The contour through $(1, 1)$ has tangent direction $(-8, 2)$. Verify that $\nabla f(1, 1)$ is orthogonal to it.
 - (d) In which direction from $(1, 1)$ does f increase fastest, and at what rate?
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Exercise 4 *An iso-performance curve — all the combinations of task difficulty d and practice p giving the same accuracy — is a contour line of the accuracy function.*

Take $A(d, p) = 1 - e^{-p/(2d)}$ with $d > 0$, $p \geq 0$.

- (a) Compute $\partial A/\partial p$. Is accuracy increasing or decreasing in practice?
- (b) Compute $\partial A/\partial d$ and give its sign. Interpret it in one sentence.

- (c) Show that the contour $A = c$ is a straight line through the origin in the (d, p) plane, and give its slope in terms of c .
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Exercise 5 *Fitting a model means finding the bottom of an error surface. With one parameter it is a curve; with two it is a bowl over the (w_0, w_1) plane.*

Let $J(w) = \sum_{i=1}^n (y_i - wx_i)^2$, a function of the single parameter w .

- (a) Compute dJ/dw .
- (b) Set it to zero and solve for w . (You have just derived least-squares regression through the origin.)
- (c) Write down one step of gradient descent, $w \leftarrow w - \eta dJ/dw$, and say in words what η controls.
- (d) Why does this J have exactly one critical point, and why is it a minimum?
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Exercise 6 Critical points.

- (a) Find all points where $\nabla f = \mathbf{0}$ for $f(x, y) = x^2 + y^2$, and say what kind of point it is.
- (b) Do the same for $f(x, y) = x^2 - y^2$. Describe the surface in words.
- (c) Do the same for $f(x, y) = e^{-(x^2+y^2)}$.
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Exercise 7 Trajectories. For each, compute $\dot{\mathbf{x}}(t)$, the speed $\|\dot{\mathbf{x}}(t)\|$, and describe the path.

- (a) $\mathbf{x}(t) = (2t, 3 - t)$;
- (b) $\mathbf{x}(t) = (t, t^2)$;
- (c) $\mathbf{x}(t) = (\cos t, \sin t)$;
- (d) $\mathbf{x}(t) = (e^{-t} \cos t, e^{-t} \sin t)$.
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Exercise 8 For $\mathbf{y}(t) = (\cos t, \sin t)$:

- (a) Show that $\mathbf{y} \cdot \dot{\mathbf{y}} = 0$ for every t .
- (b) Explain in one sentence what that says geometrically.
- (c) Give the velocity at $t = 0$ and at $t = \pi/2$, and mark both on a sketch.
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Exercise 9 *In a mouse-tracking experiment the cursor position $\mathbf{x}(t) = (x(t), y(t))$ is recorded as the participant moves toward one of two response buttons. A path that bends toward the unchosen option is evidence that the two responses competed.*

A recorded trajectory is modelled by $\mathbf{x}(t) = (t, t^2(1-t))$ for $t \in [0, 1]$, with the two buttons at $(1, 0)$ and $(1, 1)$.

- (a) Compute $\dot{\mathbf{x}}(t)$.
 - (b) Give the velocity at $t = 0$ and at $t = 1$.
 - (c) At which t is $y(t)$ largest? What does that instant represent?
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Exercise 10 Jacobians. Write $\mathbf{J}(x, y)$ for each, and evaluate it where asked.

- (a) $\mathbf{F}(x, y) = (-x + y, 2x - 3y)$. Why is $\mathbf{J}$ constant?
 - (b) $\mathbf{F}(x, y) = (x^2y, x + \sin y)$, at $(1, 0)$.
 - (c) $\mathbf{F}(x, y) = (x - x^3 - y, x - y)$, at $(0, 0)$ and at $(1, 1)$.
 - (d) For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, what is the Jacobian, and what is it usually called?
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Exercise 11

- (a) State in one line what $\mathbf{J}$ approximates: $\mathbf{F}(\mathbf{x}_0 + \boldsymbol{\delta}) \approx ?$
 - (b) Write the one-variable statement it generalises.
 - (c) Use $\mathbf{J}$ to estimate $\mathbf{F}(0.1, -0.1)$ for $\mathbf{F}(x, y) = (x - x^3 - y, x - y)$, given $\mathbf{F}(0, 0) = \mathbf{0}$.
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Exercise 12 For each matrix give $T = \text{tr } \mathbf{A}$, $D = \det \mathbf{A}$, the eigenvalues, and the type of flow of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ (stable/unstable node, saddle, spiral, centre).

- (a) $\begin{pmatrix} -2 & 0 \\ 0 & -5 \end{pmatrix}$;
 - (b) $\begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}$;
 - (c) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$;
 - (d) $\begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$;
 - (e) $\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$.
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Exercise 13

- (a) Verify that $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$ solves $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ whenever $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$.
- (b) For $\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & -4 \end{pmatrix}$ and $\mathbf{x}(0) = (2, 1)$, write $\mathbf{x}(t)$ explicitly. Which component dies first?
- (c) A system has eigenvalues $-0.1 \pm 3i$. Is it stable? Does it oscillate? At what angular speed, and with what period?
- (d) A system has eigenvalues -100 and $+0.01$. Is it stable? Explain why “mostly negative” is not enough.

Exercise 14 *A standard model of two-alternative choice: two evidence accumulators, each decaying and each inhibiting the other with strength $c > 0$.*

$$\dot{x} = -x - cy, \quad \dot{y} = -cx - y, \quad \mathbf{A} = \begin{pmatrix} -1 & -c \\ -c & -1 \end{pmatrix}.$$

- (a) $\mathbf{A}$ is symmetric. Give its two eigenvectors without computing anything, and then its two eigenvalues.
- (b) Interpret each eigenvector in words.
- (c) For which c does the difference $x - y$ grow? What does the model then do?
- (d) Take $c = 2$ and $\mathbf{x}(0) = (1.1, 0.9)$. Write $\mathbf{x}(t)$ and describe its behaviour.
- (e) Compute $D = \det \mathbf{A}$ for $c = 2$ and confirm the flow type.

Exercise 15 Nonlinear systems: nullclines, fixed points, and the Jacobian.

- (a) For $\dot{x} = y$, $\dot{y} = -x - y$: find the fixed point, write $\mathbf{J}$, give T and D , and classify.
- (b) For $\dot{x} = y - x$, $\dot{y} = x^2 - y$: find the nullclines, the fixed points, and classify each one.
- (c) For $\dot{x} = x - x^3 - y$, $\dot{y} = x - y$: find the fixed point and show that the linearisation is inconclusive. What would you do next?

Exercise 16 Putting it together. A two-variable model has a single fixed point, and its Jacobian there is $\mathbf{J} = \begin{pmatrix} -1 & \alpha \\ -1 & -1 \end{pmatrix}$, with α a parameter.

- (a) Give T and D in terms of α .
- (b) For which α is the fixed point stable?
- (c) For which α is it a spiral rather than a node?
- (d) For which α is it a saddle?